MA∗508 Time Series

Exam-like exercises for L3

(1) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{20}$, where x_k follows a Bernoulli distribution, taking the values 0 with probability 0.5 and 1 with probability 0.5, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

(2) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{20}$, where x_k follows a Bernoulli distribution, taking the values -1 with probability 1/3 and 1 with probability 2/3, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

(3) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{30}$, where x_k has the discrete distribution: -2 with probability 1/3, -1 with probability 1/3, and 1 with probability 1/3, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

(4) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{20}$, where x_k has a normal distribution with mean 0 and variance 1, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

(5) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{20}$, where x_k has the exponential distribution with parameter 1, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

(6) Sketch and describe realizations of <u>two</u> stochastic processes, x_t and y_t, that have the same one-dimensional distribution function but different (auto)covariance functions.